

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

MAR 326 — Past Questions & Materials

by Renance

Course: MAR 326

Institution: Federal University of Petroleum Resources, Effurun

Papers in this pack: 4

Levels: 300

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Paper 1 of 4 — MAR 326 Test 2021

Level 300 • Second Semester • PastQuestion

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of. Recommend four standards of measurement that can be considered in marine engineering profession and describe them. [7.5 marks]

ii). Outline three methods of measurement. What are their differences from engineering point of view? [8 marks]

MAR 326 TEST QUESTIONS

1. Marine instrument and measurement is important course in professional practice onboard vessel. Thus, knowing the basic terminologies is vital. In view of the above, define the following: (i). Calibration. (ii). Measurement. (iii). Measurand. (iv). Accuracy. (v). Precision. (vi). Sensitivity

2i). Recommend four standards of measurement that can be considered in marine engineering profession and describe them. 2ii). Outline three methods of measurement. What are their differences from the engineering point of view.

3i). In engineering field of study, instruments are classified into two. Describe them with examples. ii). Sketch block diagram of primary elements of measurement system. Justify why the block diagram should be a guide for an instrumentation company in manufacturing of digital pressure gauge for onboard vessel operations.

List the categories of such instrument and describe them with examples. [9 marks]

ii). What is the name of the instrument that can be used to determine that a marine structure has deformed. Discuss how the instrument can facilitate calculation of stress on the deformed marine structure. [8.5 marks]

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Paper 2 of 4 — MAR 326 Marine Systems

Level 300 • Second Semester • Handout

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF TECHNOLOGY
DEPARTMENT OF MARINE ENGINEERING

EXAMINATION: First Semester

SESSION: 2019/2020



COURSE CODE: MAR 326

COURSE TITLE: Marine Systems

Instrumentation

DATE: .../07/21

TIME ALLOWED: 2Hrs

INSTRUCTION: ANSWER ANY FOUR QUESTION

1a). Suppose you are one of the two "fifth engineers" onboard a marine vessel carrying out a 24 hours watch on a boiler. The data/readings obtained from a thermometer of the boiler are illustrated in the table below.

- i). What is the temperature of the boiler on the day you are on duty? [3 mark]
- ii). What is the temperature difference at 11am and 3am. Also Calculate the boiler temperature frequency. [2 marks]

Time of Watch	7am	11am	3pm	7pm	11pm	3am
Readings of Temperature of the Boiler	90°C	91°C	90°C	92°C	90°C	91°C

b). List 2 instruments that can be used for measurement of pressure, temperature, flow, liquid level, force, torque and power. [8 marks]

c). Direct and indirect methods of measurement are common terms used in marine and offshore engineering laboratories. Explain them with reference to their differences. [4.5 marks]

2a). Sketch a block diagram of primary elements of measurement system. Justify why the block diagram should be a guide for an instrumentation company in manufacturing of digital pressure gauges for onboard vessel operations. [10 marks]

b). In engineering field of study, instruments are classified into two. Describe them with examples. [5 marks]

3a. Marine instrument and measurement is an important course in professional practice onboard vessel. Thus, knowing the basic terminologies is vital. In view of the above, define the following: (i). Calibration. [1 mark] (ii). Measurement. [1 mark] (iii). Measurand. [1 mark] (iv). Accuracy. [1 mark] (v). Precision. [1 mark]

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- 6b). Name and four standards of measurement that can be considered in marine engineering profession and describe them. [7.5 marks]
- 7). Outline three methods of measurement, their uses and their differences from engineering point of view. [3 marks]
- 8a). Define an instrument with reference to marine and offshore engineering profession. [1.5 marks]
- 8b). Define the objectives of marine and offshore engineering system instrumentation. [3 marks]
- 9). Name three basic elements of a marine and offshore engineering system instrument. [3 marks]
- 10). Name two basic performance characteristics of a measuring instrument. Give two examples each. [4 marks]
- 11). Outline the advantages of a digital instrument over an analog one. [4 marks]
- 12a). Define error and describe different types of error with reference to measurement in marine and offshore engineering field of study. [9 marks]
- 12b). Describe an instrument that can be used to detect ship wrecks at the seafloor in deep waters. [3 marks]
- 6a). Name an instrument that can be used to know the angular speed of a shaft of a propeller. List the categories of such instrument and describe them with examples. [9 marks]
- 11). What is the name of the instrument that can be used to determine that a marine structure has deformed. Discuss how the instrument can facilitate calculation of stress on the aforementioned marine structure. [8.5 marks]

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Paper 3 of 4 — MAR 326 INTRODUCTION TO ERROR ANALYSIS

Level 300 • Second Semester • Handout

INTRODUCTION TO ERROR ANALYSIS

All measurements, however careful and scientific, are subject to some uncertainties. Error analysis is the study and evaluation of these uncertainties. Its three main functions are;

- It allow scientist to estimate how large uncertainties
- It helps in determining the accuracy and precision of measurement
- It is a tool that help in the reduction of uncertainties when necessary.

The challenges of estimating uncertainties and of reducing them to a level that allows a proper conclusion to be drawn can turn a dull and routine set of measurment into a truly interesting exercise.

Thus, Error analysis is the study and evaluation of uncertainty in measurement. Experience has shown that no measurement, however carefully made, can be completely free of uncertainties. Becasue the whole structure and application of science depends on measurement, the ability to evaluate these uncertainties and keep them to a minimum is crucially important.

Errros as Uncertainties

In science, the word error does not carry the usual connotations of the terms mistake or blunder. Error in a scientific measurement means the inevitable uncertainty that attends all measurements. As such, errors are not mistakes; you cannot eliminate them by being very careful. The best you can hope to do is to ensure that errors are as small as reasonably possible and to have a reliable estimate of how large they are.

Note that in this context error is used exclusively in the sense of uncertainty, and the two words are used interchangeably.

Inevitability of Uncertainty

To illustrate the inevitable occurrence of uncertainties, we have only to examine any everyday measurement carefully, for example a carpenter who must measure the height of a doorway before installing a door. As a first rough measurement, he might simply look at the doorway and estimate its height as 210 cm. This crude measurement is certainly subject to uncertainty. If pressed, the caroenter might express this uncertainty by admitting that the height could be anywhere between 205cm and 215 cm.

If he wanted a more accurate measurement, he would use a tape measure and migt find the height is 211.3 cm. This measurement is certainly more precise than his original estimate, but it is obviously still subject to some uncertainty, becasue it is impossible for him to know the height to be exactly 211.3000 cm rather than 211.3001 cm, for example.

This remaining uncertainty has many sources, which are either explicit or intrinsic to the process of measurement and intrinsic one can never be removed entirely. For example the tape is graduated in half-centimeter marks. The top of the door probably will not coincide precisely with one of the half-centimeter marks, and if it does not, the carpenter must estimate just where the top lies between two marks. Even if the top happens to coincide with one of the marks, the mark itself is perhaps a millimeter wide; so he must estimate just where the top lies within the mark. In either case, the carpenter ultimately must estimate where the top of the door lies relatively to the markings on the tape, and this necessity causes some uncertainty in the measurement.

By buying a better tape with closer and finer markings, the carpenter can reduce his uncertainty but cannot eliminate it entirely. If he becomes obsessively determined to find the height of the door with the greatest precision technically possible, he could buy an expensive laser interferometer. But even the precision of an interferometer is limited to distances of the order of the wavelength of light (about 0.5×10^{-6} meters). Although the carpenter would now be able to measure the height with fantastic precision, he still would not know the height of the doorway exactly.

Furthermore, as our carpenter strives for greater precision, he will encounter an important problem of principle. He will certainly find that the height is different in different places. Even in one place, he will find that the height varies if the temperature and humidity vary, or even if he accidentally rubs off a thin layer of dirt. In other words, he will find that there is no such thing as the height of the doorway. This kind of problem is called a problem of definition (the height of the door is not a well-defined quantity) and plays an important role in many scientific measurements.

Our carpenter's experiences illustrate a point generally found to be true, that is, that no physical quantity (a length, time, or temperature, for example) can be measured with complete certainty. With care, we may be able to reduce the uncertainties until they are extremely small, but to eliminate them entirely is impossible.

Importance of Knowing the Uncertainties

Let us now consider an example that illustrates more clearly the crucial importance of knowing how big these uncertainties are.

Suppose we are faced with a problem like the one said to have been solved by Archimedes. We are asked to find out whether a crown is made of 18-karat gold, as claimed, or a cheaper alloy. Following Archimedes, we decide to test the crown's density ρ knowing that the densities of 18-karat gold and the suspected alloy are.

$\rho_{\text{gold}} = 19.3 \text{ gram/cm}^3$ and $\rho_{\text{alloy}} = 13.8 \text{ gram/cm}^3$

If we can measure the density of the crown, we should be able (as Archimedes suggested) to decide whether the crown is really gold by comparing ρ with the known densities ρ_{gold} and ρ_{alloy}

Suppose we summon two experts in the measurement of density. The first expert, George, might make a quick measurement of ρ and report that his best estimate for ρ is 15 and that it almost certainly lies between 13.5 and 16.5 gram/cm³. Our second expert, Martha, might take a little longer and then report a best estimate of 13.9 and a probable range from 13.7 to 14.1 gram/cm³. The findings of our two experts are summarized in below.

The first point to notice about these results is that although Martha's measurement is much more precise, George's measurement is probably also correct. Each expert states a range within which he or she is confident ρ lies, and these ranges overlap; so it is perfectly possible (and even probable) that both statements are correct. But a reasonable conclusion can be drawn from the measurement of martha that the crown was not genuine, because the uncertainty is was so small and within bound.

Evidently, if the measurements are to allow a conclusion, the experimental uncertainties must not be too large. The uncertainties do not need to be extremely small, however. In this respect, our example is typical of many scientific measurements, for which uncertainties have to be reasonably small (perhaps a few percent of the measured value) but for which extreme precision is often unnecessary.

How to Report and Use Uncertainties

Best estimate $\pm$ Uncertainty

The result of measurement is to give a best estimate of the quantity and the range within which you are confident the quantity lies. For example, is the measurement of time interval gotten from a stopwatch, our uncertainty is not the difficult of reading the dial but our own unknown reaction time in starting and stopping the watch. Sometimes these kinds of uncertainty can be estimated reliably, if we can repeat the measurement several times. if a sequence of four timings gives the results (in seconds),

2.3, 2.4, 2.5, 2.4

Here, the best estimate, 2.4 s, lies at the midpoint of the estimated range of probable values, 2.3 to 2.5 s, as it has in all the examples. This relationship is obviously natural and pertains in most measurements. It allows the results of the measurement to be expressed in compact form. For example, the measurement of the time recorded in (2.1) is usually stated as follows:

measured value of time = $2.4 \pm 0.1\text{s}$

In general the result of any measurement of a quantity x is stated as follow

measured value of $x = \text{xxxxx} \pm \text{xx}$ (the standard form)

This statement means, first, that the experimenter's best estimate for the quantity concerned is the number x_{best} and second, that he or she is reasonably confident the quantity lies somewhere between $x_{\text{best}} - dx$ and $x_{\text{best}} + dx$. The number dx is called the uncertainty, or error, or margin of error in the measurement of x . For convenience, the uncertainty dx is always defined to be positive.

Questions

(a) A student measures the length of a simple pendulum and reports his best estimate as 110 mm and the range in which the length probably lies as 108 to 112 mm. Rewrite this result in the standard form.

(b) If another student reports her measurement of a current as $I = 3.05 \pm 0.03$ amps, what is the range within which I probably lies?

Measurement Uncertainties and Uncertainty Propagation

In most experiments, certain quantities are measured and then other quantities are determined from the measured data. For example, we might measure the length and width of a room, and then determine its area by multiplication. Associated with each measured quantity is a "measurement uncertainty". When a final result is determined from measured quantities, these measurement uncertainties lead to an uncertainty in the derived result. The procedure by which an experimenter determines the uncertainty in a final result from the measurement uncertainties is called "error analysis", or "propagation of uncertainty".

Types of Uncertainty

Absolute

The absolute uncertainty in a quantity is the actual amount by which the quantity is uncertain,

e.g. if $L = 6.0 \pm 0.1$ cm, the absolute uncertainty in L is 0.1 cm. Note that the absolute uncertainty of a quantity has the same units as the quantity itself.

Fractional uncertainty

The fractional uncertainty is the absolute uncertainty divided by the quantity itself, e.g. if $L = 6.0 \pm 0.1$ cm, the fractional uncertainty in L is $0.1/6.0 = 1/60$. Note that the units cancel in this division, so that fractional uncertainty is a pure number

□□

Fractional uncertainty =

|□□□□|

Percent uncertainty is fractional uncertainty expressed as a percent, i.e. fractional uncertainty multiplied by 100. If $L = 6.0 \pm 0.1$ cm, its percent uncertainty is 1.7%

Note: The word "uncertainty", by itself, normally means "absolute" uncertainty. Fractional or percent uncertainties are both called "relative" uncertainties because they relate the size of the uncertainty to the size of the result itself. While not normally used in the presentation of data or experimental results, relative uncertainties are often useful in intermediate stages of "error analysis" calculations.

Rules for Error Propagation.

The rules of error propagation refer to a situation in which we have found various quantities, $x, \dots, w$ with uncertainties $\delta x, \delta w$ and then use these values to calculation to cause an uncertainty in q as follows:

Sums and Differences:

if $q = a + \dots + b - (c + \dots + d)$

then

$$\delta q = \sqrt{(\delta a)^2 + \dots + (\delta b)^2 + (\delta c)^2 + \dots + (\delta d)^2}$$

(provided all errors are independent and random)

And

$$\delta q \leq \delta a + \dots + \delta b + \delta c + \dots + \delta d$$

Products and Quotients:

if

$$q = a \times \dots \times b$$

$$q =$$

$$q = a \times \dots \times b$$

Then

$$\delta q^2 = \delta a^2 + \dots + \delta b^2$$

$$\delta q = \sqrt{(\delta a)^2 + \dots + (\delta b)^2}$$

$$\delta q =$$

(provided all errors are independent and random)

And

$$\delta q = \delta a + \dots + \delta b + \dots + \delta c$$

$$\delta q = \delta a + \dots + \delta b + \dots + \delta c$$

$$\delta q = \delta a + \delta b + \delta c + \dots$$

Measured quantity times exact number:

if B is known exactly and

$$q = B a$$

Then

$$\delta q = B \delta a$$

$$\delta q = |B| \delta a, \text{ if } B \text{ is a constant, } =$$

$$|B| \delta a$$

Uncertainty in a Power:

If n is an exact number and

$$\square = \square \square$$

Then

11 11

$$= |\Box|$$

10 10

Uncertainty in a function of one variable:

If

$$\mathbf{h} = \mathbf{h}(\mathbf{z}) \text{ is a vector of } n \text{ hidden units}$$

11

$$\square\square = | | \square\square$$

11

You may find it easier to use the equivalent

$$R = |\sigma(\text{AAAAA} + R) - \sigma(\text{AAAAA})|$$

General formular for error propagation:

If $\phi = \phi(x, \dots, z)$ is any function of $x, \dots, z$

Then

22

11 11

$$\sigma^2 = \sqrt{(\sigma^2) + \dots + (\sigma^2)}$$

11 11

(provided all errors are independent and random)

And

00 00

$$|A| \leq |A_1| + \dots + |A_n|$$

1111

Exercise

Calculate the pressure experience by a remote operated underwater vehicle at a depth of 200m. Knowing that the depth at which it operate is measure with an uncertainty of 0.1 metre and the density of seawater is a random variable of measured value of $1024\text{kg/m}^3 \pm 5\text{kg/m}^3$. In addition state with your answer its absolute and relative uncertainties to 2.decimal place and comment on your solution.

Take $g = 9.8 \text{ m/s}^2$

$$\rho = \rho_h = 1024 \times 200 \times 9.8 = 2007040 \text{ kg}$$

Fractional uncertainty is the sum of the fractional uncertainties in ρ and h . Thus

$$\frac{1}{2} \frac{1}{h} \frac{1}{2} \frac{1}{5} \frac{1}{2} \frac{1}{0.1} \frac{1}{2}$$

$$\square\square = \sqrt{(\sqrt{}$$

$$) + () = () + () = 0.0049$$

$\square h \ 1024 \ 200$

Thus

$$P = 2007040.00 \pm 0.005 \text{ Pa}$$

$$\pm 0.0049$$

$$\text{Fractional uncertainty} = \frac{0.0049}{2007040} = 2.4414 \times 10^{-9}$$

$$\pm 2007040$$

Both uncertainties are far very small thus the measurement of pressure is of appreciable precision.

Statistical Analysis of Random Uncertainties

One of the best ways to assess the reliability of a measurement is to repeat it several times and examine the different values obtained. And not all uncertainty in measurement can be assessed by statistical analysis based on repeated measurements. For this reason, uncertainties are classified into two groups: the random uncertainties, which can be treated statistically, and the systematic uncertainties, which cannot. It's worth noting that both error propagation or statistical methods play a complementary role in error analysis. The uncertainty in measurement when you propagate error is a good measure as when you used the standard deviation of these measurements statistically.

Types of Errors

Gross Errors

This category of errors includes all the human mistakes while reading, recording and the readings. Mistakes in calculating the errors also come under this category. For example while taking the reading from the meter of the instrument he may read 21 as 31. All these types of error are come under this category. Gross errors can be avoided by using two suitable measures and they are written below:

- (i) A proper care should be taken in reading, recording the data. Also calculation of error should be done accurately.
- (ii) By increasing the number of experimenters we can reduce the gross errors. If each experimenter takes different reading at different points, then by taking average of more readings we can reduce the gross errors.

Systematic Errors

In order to understand these kinds of errors, let us categorize the systematic errors as. Systematic error are usually hard to evaluate and even to detect.

(i) Instrumental error

These errors may be due to wrong construction, calibration of the measuring instruments. These types of error may be arises due to friction or may be due to hysteresis. These types of errors also include the loading effect and misuse of the instruments. Misuse of the instruments results in the failure to adjust the zero of instruments. In order to minimize the

gross errors in measurement various correction factors must be applied and in extreme condition instrument must be re-calibrated carefully.

(ii) Environmental Errors

This type of error arises due to conditions external to the instrument.

The external condition includes temperature, pressure, humidity or it may include external magnetic field. Following are the steps that one must follow in order to minimize the environmental errors:

(a). Try to maintain the temperature and humidity of the laboratory constant by making some arrangements.

(b). Ensure that there should not be any external magnetic or electrostatic field around the instrument.

(ii). Observational Errors

As the name suggests these types of errors are due to wrong observations. The wrong observations may be due to PARALLAX. In order to minimize the PARALLAX error highly accurate meters are required, provided with mirrored scales.

Random Errors

These are uncertainties that can be revealed by repeating the measurement.

After calculating all systematic errors, it is found that there are still some errors in measurement left. These errors are known as random errors. Some of the reasons of the appearance of these errors are known but still, some reasons are unknown. Hence we cannot fully eliminate these kinds of error.

The treatment of random error is different from that of systematic error.

The statistical methods describe in the following sections gives a reliable estimate of the random uncertainties and as well as a well-define procedure for reducing them. It is expedient that a scientist should learn to anticipate the possible sources of systematic error and make sure that all systematic errors are much less than the required precision. (systematic errors should be small compared with random errors)

Engineering Applications

An engineer is designing a nylon connector to be used regularly in an auxiliary engine of a ship. The engineer is considering establishing the design specification on wall thickness at 3/32 inch but is somewhat uncertain about the effect of this decision on the connector pull-off force. If the pull-off force is too low, the connector may fail when it is installed in an engine. Eight prototype units are produced and their pull-off forces measured (in pounds):

12.6, 12.9, 13.4, 12.3, 13.6, 13.5, 12.6, 13.1, 12.5, and 13.3

Problem.

The engineer had to reach a decision point for final product specifications

The engineer employs the following

approaches

□ The dot diagram is a very useful plot for displaying a small body of data - say up to about 20 observations.

□ This plot allows us to see easily two features of the data; the location, or the middle, and the scatter or variability

A random variable, x , can be model by measured value of $x = \bar{x} + \sigma$

Where $\bar{x}$ is a constant (mean) and σ a random disturbance (deviation).

Median

The median is a measure of central tendency that divides ordered data (ascending or descending) into two equal parts, half below the median and half above. If the number of observations is even, the median is the mean between the two central values.

Considering the observations collected from the prototype engine connectors the sample median becomes

12.3, 12.5, 12.6, 12.6, 12.9, 13.1, 13.3, 13.4, 13.5, 13.6

12.9 + 13.1

$\frac{12.9 + 13.1}{2} = 13$

2

Mean

The mean or expected value of the discrete random variable X , is a weighted average of the possible values of X Denoted as μ or $E(X)$. Mathematically the sample mean of n observations denoted by

Mathematically the sample mean of n observations denoted by $x_1, x_2, \dots, x_n$ is

$\frac{x_1 + x_2 + \dots + x_n}{n} \sum_{i=1}^n x_i = 1$

$\bar{x} =$

$\frac{1}{n} \sum_{i=1}^n x_i$

Considering the observations collected from the prototype engine connectors the sample mean becomes

12.6 + 12.9 + 13.4 + 12.3 + 13.6 + 13.5 + 12.6 + 13.1 + 12.5 + 13.3

$\bar{x} =$

10

$= 12.98$

Standard Deviation

The standard deviation is a number used to tell how measurements for a group are spread out from the average (mean), or expected value. A low standard deviation means that most of the numbers are close to average. A high standard deviation means that the numbers are more spread out.

Mathematically given as

$$\sum (x_i - \bar{x})^2$$

$$\sigma = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$$

$$n-1$$

Applying this to the observations collected from the prototype engine connectors

Pull-off-

Observations

force $(x_i - \bar{x})^2$

Numbers, i

x_i

1 12.6 0.1444

2 12.9 0.0064

3 13.4 0.1764

4 12.3 0.4624

5 13.6 0.3844

6 13.5 0.2704

7 12.6 0.1444

8 13.1 0.0144

9 12.5 0.2304

10 13.3 0.1024

$$\sum = 1.9360$$

$$\sum (x_i - \bar{x})^2 = 1.9360$$

$$\sigma = \sqrt{\frac{1.9360}{10-1}} = 0.46$$

$$n-1 = 10-1 = 9$$

In summary the engineer had employ the statistical approach to bring about product design specifications. Where the pull-off-force is specified to be 12.98 ± 0.46

The Normal Distribution

The first problem in discussing measurements repeated many times is to find a way to handle and display the values obtained. One convenient method is to use a distribution or histogram. As the number of measurements approaches infinity, their distribution approaches some definite, continuous curve. When this happens, the continuous curve is called the limiting distribution.

Normal distribution, or Gauss distribution, which is the limiting distribution of results for any measurement subject to many small random errors. Once the mathematical properties of the normal distribution are understood, we can proceed to prove several important results quite easily.

Note that different types of measurements have different limiting distribution. Not all limiting distributions have the symmetric bell shape.

The limiting distribution for a measurement subject to many small random errors. The distribution is bell-shaped and centered on the true value of the

measured quantity x .

If the measurements of x are subject to many small random errors but negligible systematic errors, their distribution will be a symmetric bell-shaped curve centred on the true value X .

The Gauss function is bell-shaped and centered on $x=0$. The bell curve is wide if σ is large and narrow if σ is small

The mathematical function that describes the bell-shaped curve is called the normal distribution, or Gauss function.

$$\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$\sigma(x) = \sigma \sqrt{2\pi}$$

$$\sigma\sqrt{2\pi}$$

It's important that you become familiar with the properties of this function.

Where μ is the mean of the theoretical distribution, σ is the standard deviation, and $\pi = 3.14159 \dots$ This density function extends from $-\infty$ to $+\infty$

Acceptability of a Measured Answer

Clearly the essential point in stating any measured value is to state a range (or uncertainty) and the confidence level corresponding to that range.

The z statistic is of importance here;

Simply put, a z -score (also called a standard score) gives you an idea of how far from the mean a data point is. But more technically it's a measure of how many standard deviations below or above the population mean a raw score is.

$$z = \frac{(x - \mu)}{\sigma}$$

$$z =$$

$$z =$$

The standard score (more commonly referred to as a z -score) is a very useful statistic because it;

(a) allows us to calculate the probability of a score occurring within our normal distribution and

(b) enables us to compare two scores that are from different normal distributions.

From the table of z score we can find the probability of obtaining an answer that differs from the expected value by t or more standard deviations. This probability is

$$\text{Prob}(\text{outside } t\sigma) = 1 - \text{Prob}(\text{within } t\sigma)$$

Example:

The strengths of individual bars made by a certain manufacturing process are approximately normally distributed with mean 28.4 and standard deviation 2.95 (inappropriate units). To ensure safety, a customer requires at least 95% of the bars to be stronger than 24.0.

(a) Do the bars meet the specification?

(b) By improved manufacturing techniques the manufacturer can make the bars more uniform (that is, decrease the standard deviation). What value of the standard deviation will just meet the specification if the mean stays the same?

$$(\bar{x} - \mu) / (\sigma / \sqrt{n}) = (24.0 - 28.4) / (2.95 / \sqrt{10}) = -1.49$$

$$z = -1.49$$

$$\sigma = 2.95$$

The probability function can be read from the normal distribution table is 0.0681

The probability that the bars will be stronger than 24.0 is $1 - 0.0681 = 0.9319$ or 93.2%. Since this is less than 95%, the bars do not meet the specification.

(b) Here the σ is the unknown.

From the normal distribution table the value of z_2 for which probability function is 0.05. where probability function at $z(-1.65) = 0.0495$ and at $z(-1.64) = 0.0505$. Then z must be between -1.65 and -1.64. in this case we interpolate given

$$z_2 = -1.645$$

$$(\bar{x} - \mu) / (\sigma / \sqrt{n}) = (24.0 - 28.4) / (2.67 / \sqrt{10}) = -1.645$$

$$\sigma = 2.67$$

$$\sigma = 2.67$$

If the standard deviation can be reduced to 2.67 while keeping the mean constant, the specification will just be met.

Other Advance Distribution and Statitital Tools Includes

☐ Binomial Distribution

☐ Piosson Distribution

☐ Least-Square Fitting

☐ Covariance and Correlation

☐ Weighted Averages

Paper 4 of 4 — MAR 326 MARINE SYSTEM INSTRUMENTATION

Level 300 • Second Semester • Handout

MARINE SYSTEM INSTRUMENTATION

MAR 326

Instrumentation is a big word, with a broad and rich set of meanings. Like most words with multiple interpretations, the exact meaning is largely a function of the context in which it is used, and who is using it.

Instrumentation can be defined as the application of instruments, in the form of systems or devices, to accomplish some specific objective in terms of measurement or control, or both. Also it can be define in few as the branch of engineering that deals with measurement and control

Instrumentation has always been an integral part of technology. Development from simple level indicators, bourdon tubes, moving iron and moving coil meter etc has been rapid. Progress in electrics and electronics has led to centralised recording and display stations with associated data processing, computing and control system.

Importance of Proper Instrumentation.

The importance of using proper instrumentation electrical and mechanical instrumentation involves design, control and implementation of devices that can measure physical quantities.

There is a huge amount of thought behind the proper instrument and installation choices for each small or larger measuring point. All vessels generally use state of the art instrumentation and equipment to manage everything from the engines, steering and navigation system. When the equipment malfunctions it can have a huge impact on the operational efficiency and can cause massive downtime that can put the vessel behind schedule. A broken viscosity sensor can cause wrong engine combustion, wrong additive dosing can reduce fuel efficiency or worse; lubrication properties of various lubrication oils. A ballast that isn't filled appropriately after loading can impact vessel stability and leaks and dangerous pressure levels in fuel tanks can impact engine operation or even lead to disaster.

MEASUREMENT

Measurement is the process by which one can convert physical parameters into meaningful numbers. The measurement of a given quantity is the result of comparison between the quantity to be measured, and a definite standard. The instruments which are used for such measurements are called measuring instruments. The necessary requirements for any measuring instruments are:

- 1) The quantity to be measured should not get affected due to the instrument used.
- 2) The power consumed by the instruments for their operation should be as small

as possible.

The word measurement is used to tell us the length, the weight, the temperature, the colour or a change in one of these physical entities of a material.

Measurement provides us with means for describing the various physical and chemical parameters of materials in quantitative terms

The quantity to be measured is called measurand and the output obtained from the measurement exercise is called the result.

Measurand is classified into fundamental quantity and derived quantity.

Example of fundamental quantity is length, mass, time etc, while drive quantity is velocity, acceleration, speed, pressure etc

Two requirements which are to be satisfied to get good result from the measurement are;

- 1). The standard must be accurately known and internationally accepted
- 2). The apparatus and experimental procedure adopted for comparison must be provable.

Basic Definitions

Some important terms in 'measurement' are;

1. Instrument: A device for finding the value, or magnitude of a quantity or variable.
2. Accuracy: It is the nearness of the measured value towards the true value. ie, the measure of conformity to the true value.
3. Precision: It refers to the degree of agreement within a group of measurements or instruments. i.e. The measure of repeatability or reproducibility. Precision has two characteristics: conformity and the no. of significant figures to which measurements may be made.
4. Resolution: It is defined as the smallest change in input that can be detected by an instrument.
5. Sensitivity: It is the ratio of output signal to a change of input. Alternatively, it is the ratio of response of an instrument to a change in a measured variable.
6. True value: It is the average of the infinite no. of measurements when the average deviation tends to become zero.
7. Error: An error is the deviation from the true value of the measured variable. In order to understand the concept of errors in measurement, we should know the two terms that defines the error and these two terms are True Value and Measured Value.
8. Calibration: is the process of comparing the indication of an instrument or the value of a material measure (e.g., value of a weight or graduations of a length-measuring ruler) against values indicated by a measurement standard under specified conditions. In the process of calibration of an

instrument or material measure, the test item is either adjusted or correction factors are determined

9. Tolerance: Tolerance is the maximum allowable deviation of the value of a material measure or the indication of a measuring instrument

10. Reproducibility: The reproducibility of a measurement process is the closeness of the agreement between the results of a measurement carried out under changed conditions of measurement.

True Value: It is not possible to determine the true of quantity by experiment means. True value may be defined as the average value of an infinite number of measured values when average deviation due to various contributing factor will approach to zero.

Measured Value: It may be defined as the approximated value of true value. It can be found out by taking means of several measured readings during an experiment, by applying suitable approximations on physical conditions.

Standard Of Measurement

1. International standard

International standards are defined by international agreement and are kept at the International Bureau of Weights and Measures in Se`vres, France.

An example of an international standard is the kilogram mass (more will be said about the mass standard below). International standards are not available on a daily basis for calibration or comparison and are the basis for assigning values to other standards of the quantity.

2. Primary standard

Primary standards are maintained in national standards laboratories in countries around the world. Primary standards, representing some of the fundamental physical and electrical units, as well as some derived quantities, are independently measured and calibrated at the various national laboratories and compared against each other. Primary standards are used continually, but generally do not leave the national standards labs.

3. Secondary standard

Secondary standards are reference standards which are initially calibrated from primary standards and then used in industry and research labs on a daily basis to calibrate their working standard.

4. Working standard

These are in-house standard. They are used routinely to calibrate or check measuring instruments or material measures. A working standard is periodically compared against a secondary standard of the same quantity.

The primary reason for establishing a hierarchy scheme is to minimize the use and handling of the higher level standards and thus to preserve their values. Therefore the primary, secondary, and working standards are graded in

uncertainty, the primary standards having the best uncertainty and the working standards the worst uncertainty.

Hierarchies of measurement standards

Methods Of Measurement

1. Direct and indirect measurement
2. Primary and secondary & tertiary measurement.
3. Contact and non-contact type of measurement

1. Direct Measurement and Indirect Measurement

In direct measurement, the value of the physical parameter is determined by comparing it directly with different standards. Eg: Mass, length and time. In the Indirect Measurement, the value of the physical parameter is more generally determined by indirect comparison with the secondary standards through calibration. Example. The measurement is converting into an analogous signal which subsequently process and fed to the end device at present the result of measurement.

2. Primary and Secondary & Tertiary Measurement

Based upon the complexity of the measurement systems, the measurement is generally grouped into three categories such as Primary, Secondary and Tertiary.

□ Primary measurement

The sought value of physical parameter is determined by comparing it directly with reference standards. The required information is obtained to sense of side and touch. Example. a). Matching of two lengths is determining the length of a object with ruler. b). Estimation the temperature difference between the components of the container by inserting fingers. c). Measurement of time by counting a number of strokes of a block.

□ Secondary & Tertiary Measurement

The indirect measurements involving one transmission called secondary measurements. The indirect measurements involving two convergent are called tertiary measurements. Examples: a). The convergent of pressure into displacement by means of be allows and the convergent of force into displacement. b). Pressure measurement by manometer and the temperature measurement by mercury in glass tube thermometer. c). The measurement of static pressure by boundary tube pressure gauge is a typical example of tertiary measurement.

3. Contact and Non-contact Type of Measurements.

□ Contact type: Where the sensing element of measuring device as a contact with medium whose characteristics are being measured.

□ Non-contact type: Where the sense doesn't communicate physically with the medium. Example. The optical, radioactive and some of the

electrical/electronic measurement belong to this category